

Analysis PhD Exam, January 2003

Part I

State the following theorems.

1. Hölder's inequality.
2. The Hahn Decomposition Theorem.
3. The Fubini Theorem for positive measures μ and ν .
4. The Monotone Class Theorem.
5. The Hahn–Banach Theorem.
6. The Monotone Convergence Theorem.
7. The completeness of $L^1_E(\mu)$.
8. The theorem of uniform boundedness.

Part II

Prove one of either Theorem 3 or 7 from part I.

Part III

Solve the following problems. In problems one through four (X, Σ, μ) is a measure space and F is a Banach space.

1. Let $\mathcal{F} \subset \Sigma$ be a σ -subalgebra. Prove the existence of the conditional expectation $E(f \mid \mathcal{F})$ for $f \in L^1_F(\mu)$.
2. Let $\mathcal{R}$ be a ring generating Σ and suppose $f \in L^1_F(\mu)$. Prove, if $\int_A f \, d\mu = 0$ for each $A \in \mathcal{R}$, then $f = 0$, μ almost everywhere.
3. Suppose E (as well as F) is a Banach space and U is a bounded linear operator from E to F . Show, if $f \in L^1_E(\mu)$, then $U \circ f \in L^1_F(\mu)$ and

$$\int U \circ f \, d\mu = U \int f \, d\mu.$$

4. Suppose X is a locally compact Hausdorff space. Show, if K is a compact set and the restriction of f to K , $f|_K$, is continuous, then $f\phi_K$ is a Borel function. Here ϕ_K is the characteristic function of K .
5. Prove that $C([0, 1])$ is a closed proper subspace of $L^\infty([0, 1])$ (the L^∞ space with respect to Lebesgue measure on the interval $[0, 1]$). Explain how to view $L^1([0, 1])$ as a subspace of $L^\infty([0, 1])^*$. Is $L^1([0, 1])$ a proper subspace of $L^\infty([0, 1])^*$?